

R Charan

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Experience

Tredence Inc.

Sept 2024 – Present

Data Scientist / GenAI Engineer

• RAG-Based Chatbot

- Architected a multi-format RAG pipeline (PDF, DOCX, PPTX, Excel) with structure-aware chunking and table-level segmentation for improved retrieval precision.
- Designed hybrid search (dense + sparse) with metadata filtering on Databricks Vector Store and benchmarked GPT-4o embeddings against open-source models (BGE, E5-large).

• CDC Simple RAG Pipeline

- Built a hybrid retrieval pipeline combining vector search with SQL-based retrieval, using RRF and confidence scoring to reduce hallucination rate by ~40%.

• Root Cause Analysis (RCA) AI Agent

- Built an autonomous RCA agent using Snowflake Cortex and LLM function calling to diagnose equipment failures, cutting manual investigation time by ~50%.

Soft Designers

May 2023 – July 2023

Artificial Intelligence Intern

- Developed a real-time PPE detection system using YOLOv8n achieving 92% accuracy.
- Demonstrated the solution at Arcelor Mittal Steel Plant (Gujarat), Collins Aerospace, Hero MotoCorp, and Luminous India.
- Optimized inference pipeline using OpenCV and NumPy to achieve real-time performance (~25 FPS) on edge hardware without GPU acceleration.
- **Tech Stack:** Python, YOLOv8n, OpenCV, NumPy, Jupyter Notebook

Projects

LangGraph Multi-Agent Research Assistant

[\[GitHub Link\]](#)

- Built a multi-agent system using LangGraph where specialized agents (planner, retriever, synthesizer) collaborate to answer complex research queries with cited sources.
- Integrated web search tools and a vector store for hybrid knowledge retrieval.
- **Tech Stack:** LangGraph, LangChain, OpenAI API, FAISS, FastAPI, Python

Document Intelligence Pipeline (RAG + Structured Extraction)

[\[GitHub Link\]](#)

- Designed a production-style RAG pipeline for multi-format documents (PDF, DOCX, Excel) with metadata-aware chunking, hybrid search, and confidence-scored responses.
- Exposed pipeline via REST API (FastAPI); added evaluation using RAGAS framework.
- **Tech Stack:** LangChain, OpenAI API, ChromaDB, FastAPI, RAGAS, Python

Skills

AI / GenAI: LLMs (OpenAI, Gemini), LangChain, LangGraph, LangSmith, RAG, Agentic AI, Prompt Engineering

Vector & Embeddings: ChromaDB, Databricks Vector Store

Languages: Python, SQL, C++

Data & Cloud: Databricks, Snowflake, Azure, Docker

Backend: FastAPI, REST APIs

Education

National Institute of Technology Karnataka (NIT-K), Surathkal

July 2020 – June 2024

B.Tech – Metallurgical & Materials Engineering

CGPA: 8.13 / 10

Certifications

- Databricks Certified Generative AI Engineer Associate
- Microsoft Certified: Azure AI Engineer Associate - AI102
- Databricks Certified Machine Learning Associate